Experiments on Flame Flashback in a Quasi-2D Turbulent Wall Boundary Layer for Premixed Methane-Hydrogen-Air Mixtures

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Premixed combustion of hydrogen-rich mixtures involves the risk of flame flashback through wall boundary layers. For laminar flow conditions, the flashback mechanism is well understood and is usually correlated by a critical velocity gradient at the wall. Turbulent transport inside the boundary layer considerably increases the flashback propensity. Only tube burner setups were investigated in the past, and thus turbulent flashback limits were only derived for a fully developed Blasius wall friction profile. For turbulent flows, details of the flame propagation in proximity to the wall remain unclear. This paper presents results from a new experimental combustion rig, apt for detailed optical investigations of flame flashbacks in a turbulent wall boundary layer developing on a flat plate and being subject to an adjustable pressure gradient. Turbulent flashback limits are derived from the observed flame position inside the measurement section. The fuels investigated cover mixtures of methane, hydrogen, and air at various mixing ratios. The associated wall friction distributions are determined by Reynolds-averaged Navier-Stokes (RANS) computations of the flow inside the measurement section with fully resolved boundary layers. Consequently, the interaction between flame back pressure and incoming flow is not taken into account explicitly, in accordance with the evaluation procedure used for tube burner experiments. The results are compared with literature values, and the critical gradient concept is reviewed in light of the new data.
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1 Introduction

Current developments in the power generation sector aim toward the capture and storage of CO_2 (CCS) in order to fulfill climate protection commitments and to reduce emission costs. Precombustion capture of CO_2, one of three focused CCS process chains, yields fuels with high H_2 content exceeding 90 vol %. To date, the combustion of such H_2-rich mixtures in existing gas turbine combustors is limited to diluted diffusion-flame operation due to combustion stability constraints. Since such systems require an undesirable tight integration between gas turbine and gasification plant, lean premixed (LP) H_2 combustors are considered as an enabling technology for precombustion CCS. Recent studies [1–3] list blowout, flashback, thermoacoustic instabilities, autoignition, and emissions as operational and safety issues in modern LP combustion systems. It is interesting to mention that most of these issues do also occur in combustors running on natural gas (NG), and their understanding thus has been improved considerably during the past decades. The only exception is boundary layer flashback (BLF), which is irrelevant for NG designs but gains in importance when moving toward high H_2 contents. As a consequence, only a very limited number of investigations on BLF are available in the open literature so far. The aims of the present work are to reconsider turbulent BLF as an important instability source in H_2-fired combustors and to provide a more complete picture of the physical processes involved. It will be shown that the current understanding and data basis for turbulent BLF are neither sufficiently complete nor representative of safety-critical flame holding events.

Before turning to the turbulent situation, it is instructive to consider laminar BLF at first. Lewis and von Elbe [4] introduced a model for the physical situation at laminar BLF, which is still considered as the state of knowledge today. Figure 1 illustrates their physical picture at the onset of BLF. On the left, the wall-near region of the undisturbed boundary flow $u(y)$ is shown. Far upstream of the flame, the mixture state is determined by static temperature T and pressure p, while the wall temperature is constant at T_w. On the right-hand side of the figure, the reaction zone of the flame at the onset of flashback is depicted. The reaction zone adopts a curved shape due to the velocity gradient of the unburnt gases. At the stability limit, the burning velocity S_f at a distance δ_b from the wall equals the undisturbed flow velocity at the same height, inducing a reaction zone perpendicular to the incoming flow, which continuously ignites neighboring regions downstream. Since the flame curvature in this leading region is on the order of the flame thickness δ_f, S_f can deviate from the laminar flame speed S_L for $\mathrm{Le} \neq 1$ due to nonequidiffusion effects [5,6]. Above δ_b, the flame velocity is determined by the laminar flame speed S_L of the mixture. Close to the wall, S_f decreases due to heat loss and radical recombination, and the luminous combustion zone vanishes below the quenching distance δ_q.

Based on the situation in Fig. 1, a correlation model has been proposed, which involves the velocity gradient at the wall as a measure of the flashback limit:

$$g_c = \left.\frac{\partial u}{\partial y}\right|_{y=0} = \frac{S_f|_{y=\delta_b}}{\delta_b} \tag{1}$$

In Eq. (1), g_c is called the critical gradient, which generally depends on fuel-oxidizer kinetics, equivalence ratio Φ, T, p, T_w, the boundary layer state, and the geometry. Equation (1) induces that the distance δ_b tends to be smaller than the penetration distance; i.e., the distance from the wall at which the reaction rate starts to

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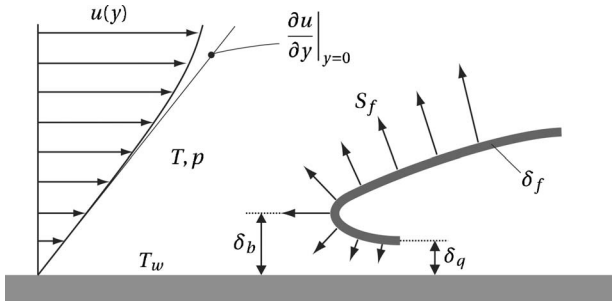

Fig. 1 2D model of flow and flame configuration at the onset of laminar BLF

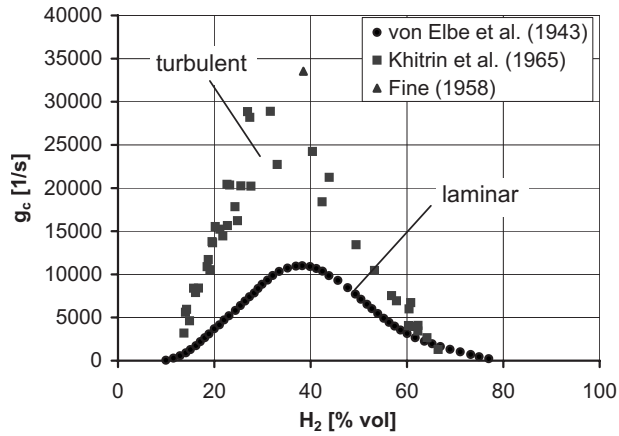

Fig. 2 Literature results for BLF in tube burners during atmospheric premixed H_2-air combustion

decrease due to the wall influence [4,7]. Putnam and Jensen [8] proposed a dimensionless correlation for laminar Hagen–Poiseuille flow based on Eq. (1), which involves two Peclet numbers:

$$\mathrm{Pe_f} = \frac{1}{8K}(\mathrm{Pe}_F)^2 \tag{2a}$$

where

$$\mathrm{Pe_f} = \frac{\bar{u}d}{a}, \quad \mathrm{Pe}_F = \frac{S_L d}{a}, \quad K = \mathrm{const} \tag{2b}$$

In Eq. (2*b*), a is the thermal diffusivity of the unburnt mixture and d is the diameter of the burner tube. Another means of nondimensionalizing Eq. (1) was proposed by Lee and T'ien [9]. For their numerical simulations of laminar tube flashback, they introduced an inverse Damköhler number,

$$\frac{1}{\mathrm{Da}_c} = \frac{g_c \delta_f}{S_L} \tag{3}$$

which has been adopted for other laminar flashback simulations [6]. For the fuels considered in the present work, numerous experimental results from laminar flashback studies were published (e.g., Refs. [10–15]). Velocity gradients were calculated from laminar flow theory and showed good agreement between different studies. In all studies mentioned, only tube or slit burner geometries with a constant cross section were investigated.

The laminar theory behind Eq. (1) was transferred unaltered to turbulent BLF. Results for atmospheric H_2-air mixtures in tube burners were published by several authors [11,12,16]. The experiments showed that the values of g_c at constant Φ increase substantially with the transition from laminar to turbulent flow. Around stoichiometry, a characteristic ratio of $g_{c,\mathrm{turb}}/g_{c,\mathrm{lam}}=3$ has been observed for H_2-air mixtures, which also seems to hold for other fuels [7]. Figure 2 summarizes literature results for turbulent critical gradients g_c as a function of the H_2 mixture fraction. To illustrate the effect of turbulence, laminar results [4] have also been included in the figure. Measurements for BLF of turbulent NG-air mixtures in tube burners are available [17], but the results are difficult to interpret since core flow velocities at the flashback point were close to the turbulent burning velocity of the mixture. In all turbulent studies mentioned so far, velocity gradients were calculated from the Blasius–Prandtl correlation for fully developed turbulent tube flow. If Eq. (1) is considered to be valid also for turbulent BLF, the assumption of $S_f=S_L$ leads to a distance δ_b, which lies within or at the border of the viscous region of the turbulent boundary layer. Although this result supports the laminar flame assumption, it does not exclude an alternative solution in the turbulent region. At the end, it remains unclear whether the increase in g_c emerges from a decrease in δ_b, an increase in S_f, or a mixture of both.

This overview on published BLF investigations reveals that in spite of early publications on experimental and theoretical basis, no substantial progress has been achieved since then to tackle the open questions regarding turbulent BLF, which definitely represents the dominating flow situation in practical burner designs. Furthermore, all experimental data available were recorded from tube burner flames, which represent a rather special case in terms of the flame-flow configuration close to the burner rim. For these reasons, a new experiment was designed, which allows the optical investigation of turbulent BLF at conditions relevant for the safety-critical assessment of boundary layers in premixed burner geometries. Instead of a tube burner arrangement, a setup was chosen where the flame is already burning inside the boundary layer of interest, and optical access was considered as a key feature to clarify the turbulent BLF mechanism using advanced optical measurement equipment. In the following, the experimental rig will be introduced, and the first measurement series of turbulent BLF limits will be presented and compared with existing tube burner data.

2 Experimental Setup

The design of the experiment follows two principles. On the one hand, a rather generic flow situation is created, which is well accessible for an analytic treatment and for measurements. On the other hand, the stable flame configuration represents the critical operation condition from a practical combustor, namely, that a flame is already present inside the boundary layer region in order to test its resistance against flashback.

2.1 Rig Structure. The experimental rig consists of flow sections for premixing, flow conditioning, the measurement section, the combustion chamber, and the exhaust duct. The premixing section allows for mixtures of air, H_2, and CH_4[2]. All gases are supplied at room temperature. A static flow mixer and the subsequent flow straightening devices provide proper large- and fine-scale mixing before the mixture enters the measurement section. An overview of measurement section and combustion chamber is shown in Fig. 3. An open air gap at the combustion chamber exit generates a constant pressure outlet condition with acoustic decoupling from the exhaust duct system.

2.2 Measurement Section. A planar asymmetric diffuser has been chosen as the basic geometry for the measurement section since diverging sections are often encountered at the exit of nonswirling as well as swirling flow burners. At the same time, high duct Reynolds numbers can be achieved, which ensures fully turbulent conditions, while flashback can still be expected due to the wall friction reducing effect of the positive pressure gradient. A

[2]NG with a mean CH_4 content of 97.63 vol % was used as a substitute.

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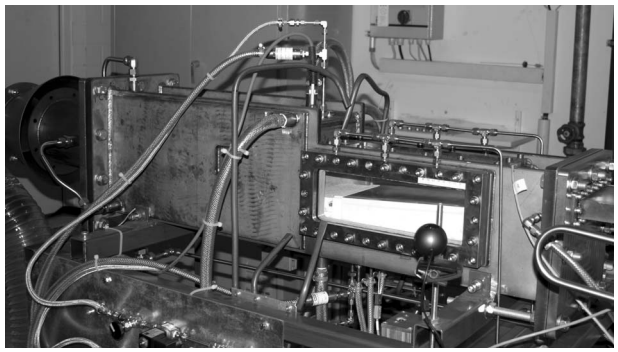

Fig. 3 Overview of measurement section and combustion chamber

further advantage of the geometry emerges from the continuously increasing wall velocity gradients along the diffuser walls in the upstream direction: At flashback, the flame propagates upstream only a finite distance until it reaches a new mean position. This behavior allows the determination of critical velocity gradients for a flame that is stabilized inside a turbulent boundary layer without any influence from auxiliary flame holders.

The flow inside the measurement section has been designed such that the flashback process only takes place in the center of the lower diffuser wall, which constitutes a quasi-2D boundary layer region with good optical access. A cross section of the diffuser arrangement is depicted in Fig. 4, with all dimensions given in millimeters. The width of the duct is constant at $w=157$ mm. The premixed gases enter the chamber from the right-hand side of the figure through an entry duct of height $h=17.5$ mm. To a short horizontal section at the inlet, a stainless steel ramp is attached with an intermediate graphite seal of approximately 0.5 mm thickness. The upper edge of the seal is positioned flush with the lower wall of the duct upstream. The ramp surface forms the lower diffuser wall with a constant inclination angle of ϕ, which can be discretely varied between 0 deg and 8 deg. The ramp surface temperature can be controlled by an impingement cooling system on the bottom side of the ramp surface, which is not shown in Fig. 4. The roughness height of the ramp surface lies within 50 μm, which results in a hydraulically smooth wall for all flow situations considered in this work. The ramp is followed flushly by a ceramic tile that ensures thermal insulation during stable operation. Optical access to the diffuser section is provided by three fused silica windows along the y- and z-directions. The window surfaces in contact with the flow are aligned with the upper and side walls, respectively, by joining two windows together using a special bonding technique. This avoids any stagnation or recirculation regions, which could lead to flame anchoring.

The premixed gases are continuously ignited by two pilot burners at the end of the asymmetric diffuser section, which thus act as flame holders for two main flames that extend into a combustion chamber of 570 mm length after a step change in cross section, as indicated on the left-hand side in Fig. 4. The pilot burners run on premixed methane and air at near-stoichiometric mixture ratios. In order to prevent blow-off, the burner flames impinge on the wall of neighboring ceramic tiles, from where the flames subsequently emerge perpendicular into the main flow. The lower pilot burner is further embedded by the trapezoidal part of a ceramic block, which forms a gap of 2 mm together with the opposite ceramic tile. The surface of the ceramic block is also inclined toward the horizontal by ϕ but is offset below the ramp plane by 4 mm. By doing so, the power of the lower pilot burner can be reduced to a minimum while still providing a stable flame to ignite the main flame. The lower pilot burner has its position still inside the diffuser section for two reasons: On the one hand, the influence of the shear layer and the recirculation zone behind the trailing edge of the ceramic block on the anchor of the lower main flame is minimized. On the other hand, the flow displacement by the lower main flame causes a contraction of streamlines above it, which induces a negative pressure gradient on the upper wall boundary layer. When flashback conditions are approached, this configuration results in a self-stabilizing system where the flame flashes back only on the lower wall of the diffuser.

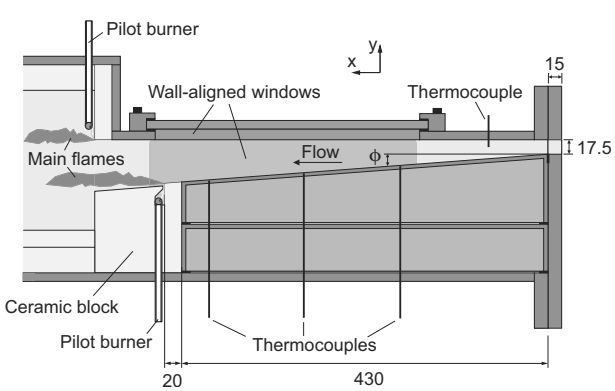

Fig. 4 Details of the measurement section

It is well known that boundary layer development in rectangular channels induces a region of reduced wall friction at the edges of the duct [18]. Therefore, according to Eq. (1), BLF is likely to occur only along the corners in this type of geometry, and, indeed, the present experiment showed this behavior as expected. However, BLF should only take place in the center region of the ramp surface, as otherwise the assumption of a quasi-2D diffuser flow would not be feasible anymore. In order to avoid corner flashback, air is locally injected into the corner boundary layers in two different ways. In the upper corners, the air enters through a slot of 2 mm width perpendicular to the main flow direction. In the lower corners, air is blown out along the main flow direction through several slots of 1 mm height and 4 mm width. During experiments, the amount of blowing air was adjusted by volume flow meters such that the sum fed to all slots did not exceed 2% of the main mass flow. The influence of the blowing air on the main flow development has been investigated quantitatively by numerical RANS mixing simulations using ANSYS CFX 12.0 as well as qualitatively by changing the amount of blowing during the running experiment. No appreciable influence, neither on the flow distribution nor on the flame shape at flashback, was observed inside the 2D region of the diffuser within the blowing range mentioned above.

2.3 Flow Conditioning. The premixed gases downstream of the static flow mixer contain substantial swirl and large-scale turbulent fluctuations. A cascade of flow straightening and settling devices downstream of the mixer is thus needed to obtain a homogeneous flow distribution, in terms of mean and turbulent quantities, at the inlet of the measurement section. For cold flow experiments, a lot of experiences exist for the design of such flow conditioning setups, which are usually composed of a wide-angle diffuser, a settling chamber of a large cross section with several honeycombs and grids, and finally a contraction toward the inlet profile. However, when premixed and preheated ignitable mixtures are investigated, a compromise has to be found between satisfactory flow conditioning and the safety aspect of avoiding large volumes with low mixture velocities. In the present setup, the conditioning section is kept at a constant cross section, and a combination of stainless-steel honeycombs, perforated plates, and wire meshes of different solidities is used as distributed eddy suppressors and pressure losses, respectively. Downstream of the last mesh, two identical rough strips of 5 mm streamwise width at the top and bottom of the channel ensure a prompt and even transition of the boundary layers.

Figure 5 shows the mean flow distribution at the combustion

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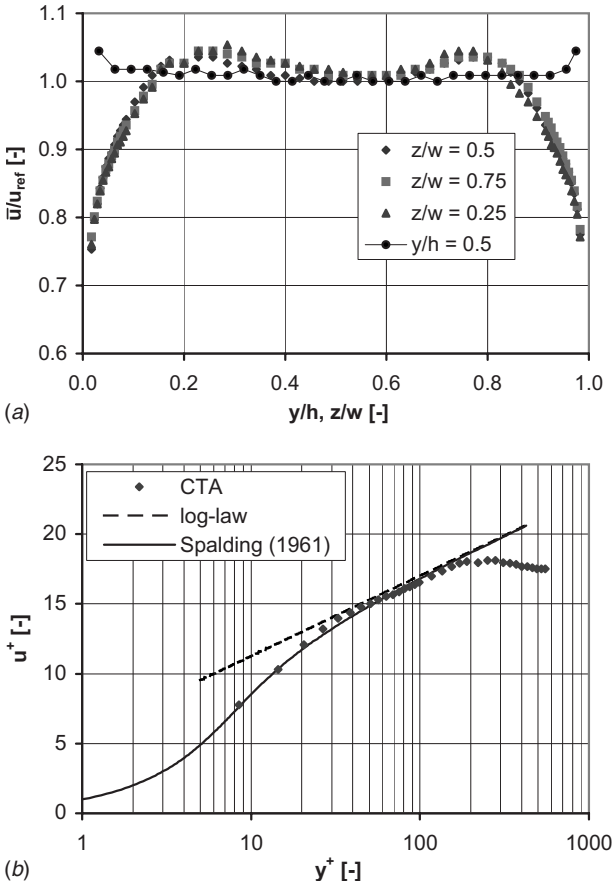

Fig. 5 Mean velocity profiles at combustion chamber inlet ($\dot{m}$=50 g/s, pure air). (*a*) Velocity profiles across full channel height and width. (*b*) Wall-near velocities at the center of the lower wall.

chamber inlet for pure air at a mass flow rate of $\dot{m}_{\mathrm{air}}=50$ g/s, which was measured by a constant temperature anemometer (CTA). The measurement plane was located 148 mm downstream of the last wire mesh.

The velocity distributions along the full channel height ($0 \leq y \leq h$) and width ($0 \leq z \leq w$) are shown in Fig. 5(*a*). The nondimensional channel height y/h and width z/w are plotted on the x-axis. On the y-axis, the nondimensionalized mean flow velocities $\bar{u}/u_{\mathrm{ref}}$ are shown. The mean velocity at the center of the duct was chosen as u_{ref}. The velocity distribution along the channel width at half channel height $y/h=0.5$ shows good uniformity. Only close to the channel sidewalls, the velocities rise above the center value. This fact can be explained by the flow distribution properties of fine wire meshes, which produce a distinct velocity overshoot at the edge of the boundary layer due to a minimum of the pressure loss coefficient in this region [19]. The same mechanism is observed for the mean velocity profiles along the channel height at three lateral positions $z/w=0.25$, 0.5, and 0.75. While the profiles again show a good uniformity within $\pm 1.5\%$, there exists a symmetric peak of velocities above the center value at $y/h=0.25$ and $y/h=0.75$. The boundary layer thickness amounts to approximately 3 mm at the top and bottom of the channel. The measurements did not extend to the near wall region of the channel sidewalls, as can be seen from the different scaling of the x-axis. However, the boundary layer thickness there is probably below 3 mm since the sidewall regions lack the momentum loss from the rough strips upstream.

According to Ref. [19], the wall friction produced by such mesh wake profiles tends to lie above the value for an undisturbed turbulent boundary layer at the same position. In order to check whether this is also true for the present flow, the CTA results at the center of the lower wall were transformed to the wall coordinates,

$$y^+ = \frac{y}{\nu} u_\tau \tag{4a}$$

$$u^+ = \frac{\bar{u}}{u_\tau} \tag{4b}$$

In Eq. (4*a*), ν is the kinematic viscosity of the mixture and u_τ is the shear stress velocity, calculated from the wall shear τ_w and the mixture density ρ by $u_\tau=\sqrt{\tau_w/\rho}$. Figure 5(*b*) shows the result from the coordinate transformation. The equation for the logarithmic region of a fully turbulent boundary layer,

$$u^+ = \frac{1}{\kappa}\ln(y^+) + B \tag{5}$$

as well the law of the wall derived by Spalding [20] have been included in the figure. For both correlations, $\kappa=0.4$ and $B=5.5$ were assumed. The wall shear stress τ_w was adjusted until a good match between the CTA results and the Spalding correlation was achieved. The wall shear stress determined in this way amounts to $\tau_w=1.02$ Pa, which is close to numerical results for fully developed channel flow at the same mass flow and density (see the next section for simulation details). It can be concluded that the wall-near region of the boundary layer at the combustion chamber inlet has the self-similar structure of a fully turbulent boundary layer, and the wall shear stress is comparable to the results for fully developed turbulent channel flow.

2.4 Instrumentation. Mixture ratios for each component as well as the pilot burner fuel mass flow are controlled by separate thermal mass flow controllers with a nominal accuracy of 0.8% of the measured value. The mixture inlet temperature and the center temperature of the ramp surface plate at three positions are measured by thermocouples. For the optical measurements presented in this work, an intensified charge-coupled device (ICCD) camera (Hamamatsu C4336-02) with a nominal resolution of 720 $\times$480 pixel and a frame rate of 30 Hz was used. An interference filter with a transmission range of 300–350 nm in combination with a UV lens was used to confine the main part of the captured signal to the radiation of the OH^* radical.

3 Measurement Procedure

For the determination of critical velocity gradients g_c at turbulent BLF, a combination of experimental and numerical results was used. First, the mean position of the flame tip at flashback was calculated from a time series of instantaneous flame images for a certain inlet mass flow and equivalence ratio. As described before, these images were taken at the pseudostationary new equilibrium position of the flame after it had flashed back from its original position above the pilot burner. Second, the respective mean wall friction distribution along the centerline of the ramp surface was extracted from a 2D RANS simulation of the flow inside the measurement section, and the value of g_c at the mean flame tip position was evaluated. This procedure will be explained in more detail in the following paragraphs.

3.1 Mean Flame Tip Position at BLF. To give an illustration of the observed flame configuration, Fig. 6 shows the flame shape and position inside the measurement section at BLF for a lean H_2-air mixture at $\dot{m}_{\mathrm{air}}=100$ g/s and $\Phi=0.29$. The image represents an overlay of the flame chemiluminescence captured with the ICCD camera with an image of the measurement section at ambient light. Since the depth of focus was large compared with the channel width, the flame image represents a line-of-sight integration with the optical axis tangential to the ramp surface plane. The flame exhibits the expected shape at BLF on the lower dif-

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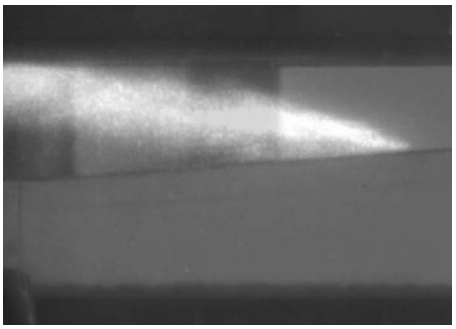

Fig. 6 Instantaneous H_2-air flame shape and position at BLF ($\dot{m}_{air}$=100 g/s, Φ=0.29)

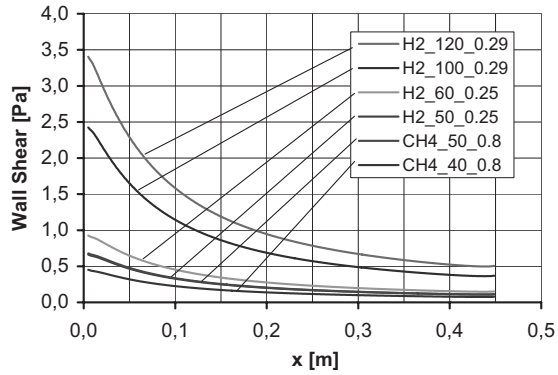

Fig. 7 Numerical results for wall shear stress along the lower diffuser wall

fuser wall, including a strongly curved leading edge zone and an angled, stretched tail toward the inner channel region (see Fig. 1). A top view on the leading flame front, which is not shown in Fig. 6, reveals a satisfactorily even mean lateral distribution in the center region of the ramp. At the sidewalls of the channel, the reaction is quenched since the blowing air produces equivalence ratios below the lower explosion limit of the mixture. Since the flame tip is not stationary during BLF, but is moving back and forth at an oscillation frequency of approximately 1–2 Hz due to turbulent fluctuations, a mean flame tip position has been calculated from the experimental data. For each operation point, single images from two measurement series (30 s duration each) were binarized with regard to the flame region, and a time averaged image normalized between 0 and 1 was produced. Along a line parallel to the ramp surface, which intersected the mean flame tip region, and approaching this region from upstream, the point of 0.5 intensity was taken as the mean flame tip position at BLF.

3.2 Numerical Simulation of the Diffuser Flow. The wall friction distribution for each investigated combination of inlet mass flow and mixture composition was simulated using the ANSYS CFX 12.0 code. All simulations were made for cold mixture flow without combustion. By doing so, the data evaluation is analogous to the procedure used for tube burner experiments in the literature, where wall gradients were calculated from tube friction correlations, which are only valid for the nonreactive case. A hexahedral 2D mesh with 0.35×10^6 nodes was used to represent the lateral center region of the measurement section and the combustion chamber. The fluid was modeled as a constant property mixture of ideal gases with the dynamic viscosity taken from pure air at 300 K and 1 bar. The domain additionally included a straight duct of height h and a length of $50h$ in order to establish a fully developed channel profile before the beginning of the diffuser section. As mentioned before, the mean inlet velocity profile in the experiments generated a wall velocity gradient, which is close to the fully developed channel flow case. Therefore, the numerical simulation is expected to be a good representation of the wall friction on the lower diffuser wall during the experiments. The boundary layers along the diffuser walls were fully resolved during all simulations, with a maximum value of $y^+_{max}<0.4$ at the center of the wall cells. It was ensured that the laminar sublayer contained more than three grid points and that the whole boundary layer contained more than 30 grid points. The pilot burner mass flows were neglected due to their vanishing influence on the flow distribution. A constant average pressure outlet was set at the outlet of the combustion chamber, while the domain was bounded by symmetry planes at both sides. Turbulence modeling was provided by the shear stress transport (SST) two-equation model with the default values for all model constants. This choice was based on the outcomes of several test cases for wall friction simulations [21,22], which consistently revealed very good prediction capabilities of $k-\omega$ based RANS models for turbulent wall friction at mild to moderate adverse pressure gradients. Figure 7 shows the numerical results for the wall shear distribution on the lower wall of the diffuser. On the x-axis, the horizontal coordinate is shown, where the point $x=0$ is located at the start of the diffuser section. The curves exhibit the expected shape with a continuous decrease in wall shear in the downstream direction. The maximum of the wall shear gradient is located at the beginning of the diffuser, which is caused by a maximum of the streamwise pressure gradient in this region.

4 Results and Discussion

According to the experimental procedure described above, BLF experiments were conducted and mean critical gradients $\bar{g}_c$ were determined for several mixtures of CH_4, H_2, and air. The inclination angle ϕ of the ramp surface was kept constant at 4 deg, which resulted in a diffuser flow with no appreciable stall, as predicted from corresponding diffuser flow charts [23].

The time averaged images for the determination of the mean flame tip position at turbulent BLF are shown in Fig. 8. As for Fig. 6, the geometry of the measurement section has been used as a background to make the images more instructive. The images show very similar shapes, and it can be seen that there is a unique mean flame position when approaching the flame tip region from upstream. It should be emphasized that the broad bright zone behind the leading tip of the flame does not indicate a high residence probability of the flame tip in this region. It rather represents the time average of the chemiluminescence from the tail of the flame.

The quantitative results from the combined experimental and numerical data processing are summarized in Table 1. In each line, the fuel as well as the air mass flow rate $\dot{m}_{air}$ are listed, followed by the equivalence ratio Φ_{BLF} at which the position of the flame at flashback was recorded. The mean critical gradient $\bar{g}_c$ is given as extracted from the respective numerical wall friction simulation. The last entry, $\bar{g}_{c,t^*=0.5-1}$, is the mean critical gradient as calculated from the second half of each data set. This variable is a measure of the influence of wall heating during the first half of the experiments ($t^*=0-0.5$) on the further flashback position. By comparing the results for $\bar{g}_c$ and $\bar{g}_{c,t^*=0.5-1}$ in each line, it is concluded that the heating of the ramp during the experimental time had no significant effect on the determined mean flashback positions.

The obtained results for the mean critical gradients were surprising. For all mixtures, the critical gradients lie substantially above the tube values reported in the literature. The results for CH_4-air mixtures can be compared with the turbulent flashback data published by Grumer [17], which indicates critical gradients below g_c=2000 1/s for Φ=0.8. This value is exceeded by the present results by a factor of about 3. The results for H_2-air mixtures can be directly compared with Fig. 2. It can be seen that at such lean mixtures (9.5 vol % H_2 for Φ=0.25 and 10.71 vol % H_2 for Φ=0.29), flashback in tube burners was observed at laminar

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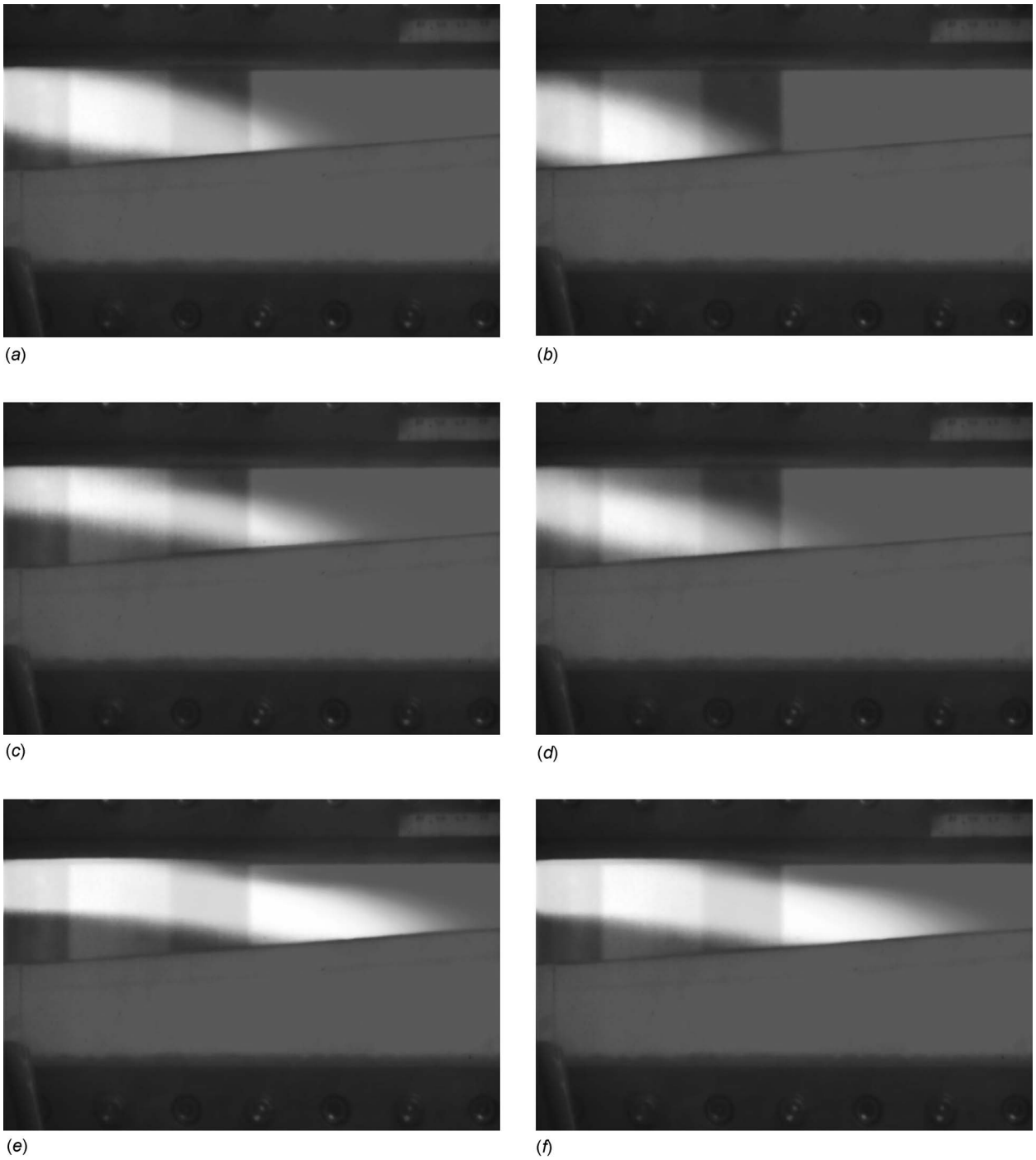

Fig. 8 Time averaged flame images for the determination of mean flame tip position at turbulent BLF. (*a*) CH_4, $\dot{m}_{air}$ =40 g/s, and Φ=0.8. (*b*) CH_4, $\dot{m}_{air}$=50 g/s, and Φ=0.8. (*c*) H_2, $\dot{m}_{air}$=50 g/s, and Φ=0.25. (*d*) H_2, $\dot{m}_{air}$=60 g/s, and Φ =0.25. (*e*) H_2, $\dot{m}_{air}$=100 g/s, and Φ=0.29. (*f*) H_2, $\dot{m}_{air}$=120 g/s, and Φ=0.29.

conditions only at very low critical gradients around g_c =100 1/s. The present results exceed this laminar value by two orders of magnitude.

The reasons for the observed deviations of the present results from the literature are probably twofold: (1) The pressure rise upstream of the reaction zone acts more strongly on the incoming boundary layer flow, and (2) the boundary layer resistance to flashback is deteriorated by the positive pressure gradient acting on it.

Looking at the first point, the configuration between flame and incoming flow at conditions approaching flashback is not the same

Table 1 Critical gradients at BLF for different fuel-air mixtures at a diffuser angle of 4 deg

Fuel	$\dot{m}_{air}$ (g/s)	Φ_{BLF}	$\bar{g}_c$ (1/s)	$\bar{g}_{c,t^*=0.5-1}$ (1/s)
CH_4	40	0.8	5329	5336
CH_4	50	0.8	6952	6952
H_2	50	0.25	8098	8067
H_2	60	0.25	10,047	10,121
H_2	100	0.29	31,899	31,798
H_2	120	0.29	42,576	42,786

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for tube burners and the experiment presented here. At the exit of a tube burner, the flame is stabilized inside the shear layer formed in the wake of the burner rim. Consequently, the backpressure of the flame is not limited to the boundary layer region upstream of the burner rim but is also distributed to regions directly above and toward the outward side of the burner rim. In the present experiment, the lower main flame always stays attached to the edge above the pilot burner exit, where its pressure rise acts directly against the incoming boundary layer. The induced local deceleration of the near-wall flow is likely to support the upstream propagation of the flame dynamically, depending on the dynamic pressure profile of the boundary layer and the pressure rise in front of the flame.

Regarding the second argument, it is instructive to compare values from Table 1 for $\bar{g}_c$ at identical Φ but different $\dot{m}_{\mathrm{air}}$. It becomes obvious that for all three data pairs, which span over both fuels and a wide range of $\bar{g}_c$, the critical gradients for high $\dot{m}_{\mathrm{air}}$ lie consistently above the data for lower $\dot{m}_{\mathrm{air}}$. In other words, if the mean position at BLF lies more downstream from the diffuser inlet, as it is the case for higher $\dot{m}_{\mathrm{air}}$ at a given Φ, then the boundary layer is more susceptible to flashback compared with positions closer to the inlet. An obvious explanation for this observation is the integral distance along which the positive pressure gradient has acted on the boundary layer. This is an interesting observation since turbulent boundary layers subject to an adverse pressure gradient only show deviations from the universal nondimensional profile in the outer layer, which has also been called the wake region (e.g., Ref. [24]). However, according to the classical theory explained in the introduction, turbulent BLF is determined only by the flow distribution inside the laminar sublayer, which should stay the same at different positions along the ramp.

5 Conclusions

An experimental setup has been presented, which allows the optical investigation of turbulent BLF during premixed combustion of CH_4–H_2-air mixtures. The flame holding has been chosen such that it represents the critical situation in gas turbine combustors, namely, that a wall boundary layer should be resistant against flashback even if a flame is directly located inside of it. As a first series of measurements from the new rig, mean flame tip positions during BLF were recorded for different mixture compositions and inlet mass flows. The flow was subject to an adverse pressure gradient imposed by an asymmetric diffuser at an opening angle of 4 deg. The results were expressed as mean critical velocity gradients according to the classical flashback theory using numerical RANS simulations for the determination of the wall shear distributions inside the measurement section. The results deviate substantially from the published data for tube burner flashbacks, indicating that the investigated combination of the safety-critical flame position and the influence of an adverse pressure gradient upon the boundary layer structure leads to a completely different interaction between flame and flow than expected. In light of these findings, the concept of a critical velocity gradient as a universal measure of flashback propensity in turbulent boundary layers seems not to deliver realistic results and has to be replaced by a model which also takes into account the outer regions of the boundary layer. Future work will be concentrated on the influence of the pressure gradient by varying the diffuser opening angle, on mixture preheating, on the interaction between the flame backpressure and the incoming boundary layer as well as the detailed flame propagation mechanism inside the boundary layer.

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